

Linux basic commands

1. User Management & File Inspection

Write a Bash script that creates a new user, sets a password, and verifies the user's entry in /etc/passwd using grep. What command would you use to check the file type before opening it?

PROGRAM:

```
#!/bin/bash

#Giving value to variable
USERNAME=$1
PASSWORD=$2

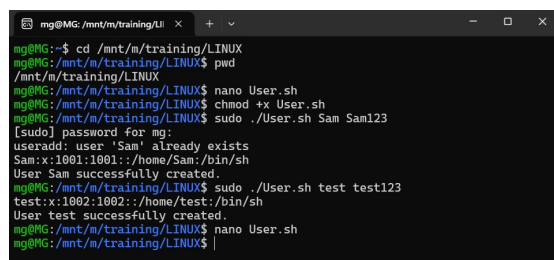
#New username created and added
sudo useradd $USERNAME

#Set user Password
echo "$USERNAME:$PASSWORD"|sudo chpasswd

#checking the password
grep "^$USERNAME:" /etc/passwd

#file open checking
if [ $? -eq 0 ]
then
    echo "User $USERNAME successfully created."
else
    echo "User creation failed."
Fi
```

OUTPUT

A terminal window titled 'mg@MG: /mnt/m/training/LI' showing the execution of a Bash script. The script creates a user named 'Sam' with password 'Sam123' and verifies its entry in /etc/passwd. The output shows that the user was successfully created and the password was set correctly. The terminal text is as follows:

```
mg@MG:~$ cd /mnt/m/training/LINUX
mg@MG:/mnt/m/training/LINUX$ pwd
/mnt/m/training/LINUX
mg@MG:/mnt/m/training/LINUX$ nano User.sh
mg@MG:/mnt/m/training/LINUX$ chmod +x User.sh
mg@MG:/mnt/m/training/LINUX$ sudo ./User.sh Sam Sam123
[sudo] password for mg:
useradd: user 'Sam' already exists
Sam:x:1001:1001::/home/Sam:/bin/sh
User Sam successfully created.
mg@MG:/mnt/m/training/LINUX$ sudo ./User.sh test test123
test:x:1002:1002::/home/test:/bin/sh
User test successfully created.
mg@MG:/mnt/m/training/LINUX$ nano User.sh
mg@MG:/mnt/m/training/LINUX$
```

2. Environment Variables and Shell Configuration

Modify `.bash_profile` to include a custom greeting and a new environment variable. How would you ensure the variable is available in child shells?

PROGRAM:

```
echo "Welcome MG! Linux environment ready."
```

```
export TRAINING_PATH="/mnt/m/training/LINUX"
```

OUTPUT:

```
mg@MG:/mnt/m/training/LINUX$ nano User.sh
mg@MG:/mnt/m/training/LINUX$ nano ~/.bash_profile
mg@MG:/mnt/m/training/LINUX$ source ~/.bash_profile
Welcome MG! Linux environment ready.
mg@MG:/mnt/m/training/LINUX$ echo $TRAINING_PATH
/mnt/m/training/LINUX
mg@MG:/mnt/m/training/LINUX$ export TRAINING_PATH="/mnt/m/training/LINUX"
mg@MG:/mnt/m/training/LINUX$ bash
echo $TRAINING_PATH
mg@MG:/mnt/m/training/LINUX$ bash
mg@MG:/mnt/m/training/LINUX$ echo $TRAINING_PATH
/mnt/m/training/LINUX
mg@MG:/mnt/m/training/LINUX$ |
```

3. Redirection Operators

Create a script that collects system information using `uname -a` and `ip a`, and saves it to a file. Ensure that the output is appended, not overwritten. How would you redirect both `STDOUT` and `STDERR` to the same file?

PROGRAM:

```
#!/bin/bash
```

```
echo "System Information" >> system_info.txt
uname -a >> system_info.txt 2>&1
```

```
echo "Network Information" >> system_info.txt
ip a >> system_info.txt 2>&1
```

OUTPUT:

```
mg@MG:/mnt/m/training/Li  X  +  v
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ nano redirect.sh
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ >>
bash: syntax error near unexpected token `newline'
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ uname -a >> system_info.txt
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ >> file 2>&1
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ ip a >> system_info.txt 2>&1
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ chmod +x sysinfo.sh
./sysinfo.sh
chmod: cannot access 'sysinfo.sh': No such file or directory
bash: ./sysinfo.sh: No such file or directory
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ chmod +x sysinfo.sh
chmod: cannot access 'sysinfo.sh': No such file or directory
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ chmod +x Redirect.sh
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ ./Redirect.sh
mg@MG:/mnt/m/training/LINUX/SHELL SCRIPTING$ cat system_info.txt
Linux MG 5.15.107.4-microsoft-standard-WSL2 #1 SMP Tue Nov 5 00:21:55 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:c2:c6:e4 brd ff:ff:ff:ff:ff:ff
    inet 172.24.238.60/20 brd 172.24.239.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5d:ffc2:c6e4/64 scope link
        valid_lft forever preferred_lft forever
-----
System Information
Linux MG 5.15.107.4-microsoft-standard-WSL2 #1 SMP Tue Nov 5 00:21:55 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux
-----
Network Information
-----
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:c2:c6:e4 brd ff:ff:ff:ff:ff:ff
    inet 172.24.238.60/20 brd 172.24.239.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5d:ffc2:c6e4/64 scope link
        valid_lft forever preferred_lft forever
```

4. Positional Parameters

Create a script that accepts four arguments: profession, age, city, and name. Use positional parameters to print a formatted sentence using these inputs.

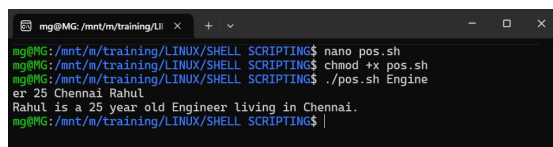
PROGRAM:

```
#!/bin/bash

profession=$1
age=$2
city=$3
name=$4

echo "$name is a $age year old $profession living in $city."
```

OUTPUT:



```
mg@MG: /mnt/m/training/LI x + v
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ nano pos.sh
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ chmod +x pos.sh
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ ./pos.sh Engine
er 25 Chennai Rahul
Rahul is a 25 year old Engineer living in Chennai.
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$
```

5. Control Operators

Write a script that checks if a file exists and prints a message using &&. Then modify it to print a different message if the file does not exist using ||.

PROGRAM:

```
#!/bin/bash

FILE=$1

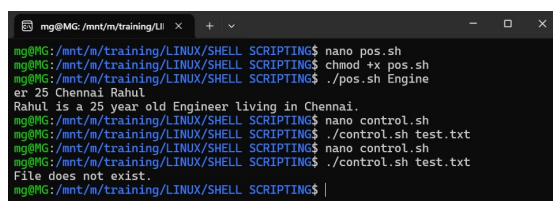
[ -f "$FILE" ] && echo "File exists."

#!/bin/bash

FILE=$1

[ -f "$FILE" ] && echo "File exists." || echo "File does not exist."
```

OUTPUT:



```
mg@MG: /mnt/m/training/LI x + v
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ nano pos.sh
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ chmod +x pos.sh
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ ./pos.sh Engine
er 25 Chennai Rahul
Rahul is a 25 year old Engineer living in Chennai.
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ nano control.sh
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ ./control.sh test.txt
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ nano control.sh
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$ ./control.sh test.txt
File does not exist.
mg@MG: /mnt/m/training/LINUX/SHELL SCRIPTING$
```